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Simulation of Optical Emission Spectra of Cu-Sn-Zn Alloy Plasmas for Laser-Induced Breakdown Spectroscopy Applications

M S Suryan Sivasdas¹, Lekha Mary John¹ and K K Anoop^{1,*}

¹Department of Physics, Cochin University of Science and Technology, Kochi-682 022

E-mail: *anoopkk@cusat.ac.in

Abstract. Atomic emission spectroscopy (AES) is a potential tool for qualitative and quantitative elemental analysis of multi-element materials. The emission spectrum is composed of the characteristic lines of the constituent species, and their intensities are proportional to the population of their energy states and transition probability. We simulated the emission spectrum of copper-tin-zinc alloys under typical laser-induced plasma conditions (e.g. $T_e = 1$ eV and $N_e = 10^{17} \text{ cm}^{-3}$). Saha LTE equation determines the population of neutral and ionised species for each of these constituent elements. The Boltzmann distribution is used to calculate the strength of their probable transitions, and each line is given a broadened Gaussian profile. Their sum will approximate an ideal plasma spectrum of the alloy. Spectra are generated for varying temperatures, electron density, and compositions and show variations based on these parameters. This simulation provides a simple and convenient way to perform laser-induced breakdown spectroscopy (LIBS) applications on any alloy sample.

1. Introduction

Atomic emission spectroscopy (AES), is a widely used tool for qualitative and quantitative analysis of multi-element samples like alloys. In AES, samples are excited to a high-temperature plasma state, through electrical discharge or laser ablation. This plasma contains its constituent species' in their excited states and they will emit radiation upon returning to their ground state. This emitted light corresponds to the characteristic emission of lines of these species' and constitute the plasma emission spectrum. Intensity of these lines are proportional to the population of the excited energy state and transition probability. This makes the emission spectra useful for the qualitative and quantitative analysis of these samples. Laser-Induced Plasma (LIP) is a transient micro plasma that shows spatial and temporal inhomogeneity. Typical laser produce plasma contains 10^{17} to 10^{19} electrons per unit volume (cm^{-3}), and plasma temperature ranges from 6000 K to 20000 K[1]. And they mainly depend on laser pulse characteristics (pulse width, pulse energy, and wavelength), ambient conditions (pressure), and sample (physical and chemical properties). Electron number density and plasma temperature vary in the different temporal and spatial windows; due to this nature, a complete thermodynamic equilibrium (CTE) state cannot be explained for laser-produced plasma; therefore, we use the Local Thermodynamic Equilibrium (LTE) assumption[2]. To give a correct interpretation of plasma spectra and for the meaningful application of theoretical Boltzmann–Maxwell and Saha–Eggert expressions, the existence of LTE is essential[3].

In this work, we simulated emission spectrum of some copper-tin-zinc alloys under typical laser-induced plasma conditions. We have used Saha's LTE equation and Boltzmann statistics to estimate the intensity of possible transitions and gave a broadened Gaussian profile to each line.

2. Simulation Details

In this section, we describe the steps in the plasma spectrum simulation in detail. There are three input variables, which specifies the condition of the plasma. These are the plasma temperature (T_e) number density of electrons in the plasma (n_e) and the fractional composition of each constituent element in the plasma. This composition is expressed as *weight fractions*, (denoted as w_X for an element X and $\sum_X w_X = 1$). For calculations, we need the number *fraction*, which can be computed from the



weight fraction, knowing the atomic mass of the element as and the proportionality is set by the normalization. Besides these inputs, we need a database of all possible transitions of the constituent elements and related information like transition probability, level energies and corresponding degeneracies. These data were taken from NIST atomic spectroscopy database[4] and Kurucz database[5].

2.1. Step 1: Estimating Contributions from Different Species

In the plasma, an element X can exist in its neutral state or any of its ionized states. All these states will have their contribution to the plasma spectrum, based on their abundance. So, the first step is to quantify the abundance of each ionized species. For this, we assume *the plasma to be in LTE*. Then, we can use *Saha's ionization equation*, which relate the abundance of a certain ionization state to the next higher state of the same element in a plasma at LTE, at some temperature[6,7]. It is given by

$$\frac{n_{X,i+1}}{n_{X,i}} = 6.009 \times 10^{21} \frac{T_e^{3/2}}{n_e} \cdot \frac{U_{X,i+1}(T_e)}{U_{X,i}(T_e)} \cdot \exp\left(-\frac{I_{X,i}}{T_e}\right) \quad (1)$$

Where $n_{X,i+1}$ is the number fraction, $U_{X,i}$ is the partition function and $I_{X,i}$ is the ionization energy of the i -th ionization state of element X. (In this section, we adopt the notation X^{i+} to denote the i -th ionization state of an element X. The 0-th ionization state is the neutral species.) With these ratios and the relation $\sum_i n_{X,i} = n_X$, we can estimate the fractional contribution of each state of a particular element. In our simulation, we only considered the first two ionization states of each element and their abundances can be found to be.

$$n_{X,0} = \frac{n_X}{1+y} \text{ and } n_{X,1} = yn_{X,0}, \quad \text{where } y = \frac{n_{X,1}}{n_{X,0}} \quad (2)$$

2.2. Step 2: Estimation of Peak Line Intensities

As mentioned earlier, each constituent species in the plasma has their own contribution to the spectrum. Each of their characteristic transitions appears in the spectrum, with intensities depending on their population and the probability of that transition. Let us consider the species X^{i+} . If it has a transition $u \rightarrow l$, with a probability A_{ul} , resulting in a line at wavelength λ_{ul} , then its intensity can be estimated as[6].

$$I_{ul} := I(\lambda_{ul}) = \frac{1}{4\pi} n_{ul} A_{ul} \epsilon_{ul} \quad (3)$$

Here, $\epsilon_{ul} = E_u - E_l$ is the energy of the emitted photon, which is the difference between the upper and lower-level energies and n_u is the fractional population of the upper level u of that species. The latter depends on the specific statistics the particles (here, atoms and ions) follows. For a fairly high temperature plasma, one could use the *Boltzmann statistics*, which gives the population to be

$$n_u(T) = \frac{n}{U(T)} g_u \exp\left(\frac{-E_u}{T}\right), \quad \text{where } U(T) = \sum_j g_j \exp\left(\frac{-E_j}{T}\right) \quad (4)$$

Here n is the total population of the species and g_u is the degeneracy for energy level E_u . Required spectroscopic data are collected from NIST Atomic spectra database[4] and Kurucz atomic line list[5]. With this, intensities of all the available lines at a specific set of temperature and electron density are pre-computed so that spectra can be generated at various wavelength ranges as desired.

2.3. Step 3: Generating the Final Spectrum

Last step in the simulation is to generate the final spectrum in the desired wavelength range. For this, we give each line a broadened shape and sum over all available lines to produce the output spectrum.

Known mechanisms of spectral broadening are the *Doppler* and *Stark* mechanisms. But, for simplicity, we only consider a broadening that is attributed to the *resolving power* of the spectrometer, given by $R = \lambda_0/w$. Here, w is the *full width at half maximum* (FWHM) of the broadened line and λ_0 is its central wavelength [8]. We have chosen a Gaussian line shape function for each line, given by (see, for example, [9]).

$$I(\lambda) = I_0 \exp \left[-\frac{1}{2} \left(\frac{\lambda - \lambda_0}{s} \right)^2 \right] \quad (5)$$

Where $w = 2s\sqrt{2\ln 2}$ and I_0 is the peak intensity of the line, as given by Eqn. (3). We have also assumed the plasma to be optically thin, so that effects such as self-absorption can be avoided. Final spectrum will be the sum of all the available line profiles. The result will resemble the plasma spectrum in ideal conditions.

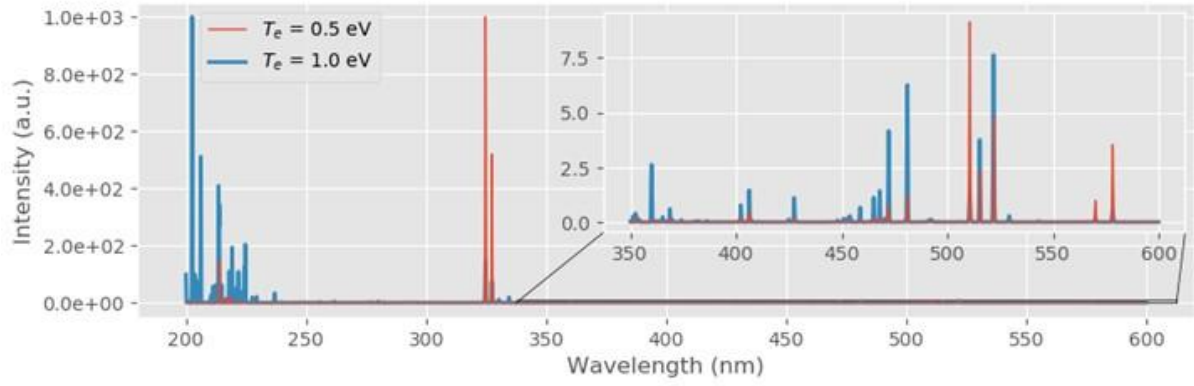
3. Results and Discussion

Using the method described in the former section, we have simulated spectra for three different alloys, namely *brass* (65% copper and 35% zinc by weight), *bronze* (88% copper and 12% tin) and *admiralty brass* (69% copper, 30% zinc and 1% tin). Simulations are done at temperatures of 0.5 and 1 eV, with an electron density of 10^{17} cm^{-3} . Simulated spectra are presented in Fig. (1a), (1b) and (1c).

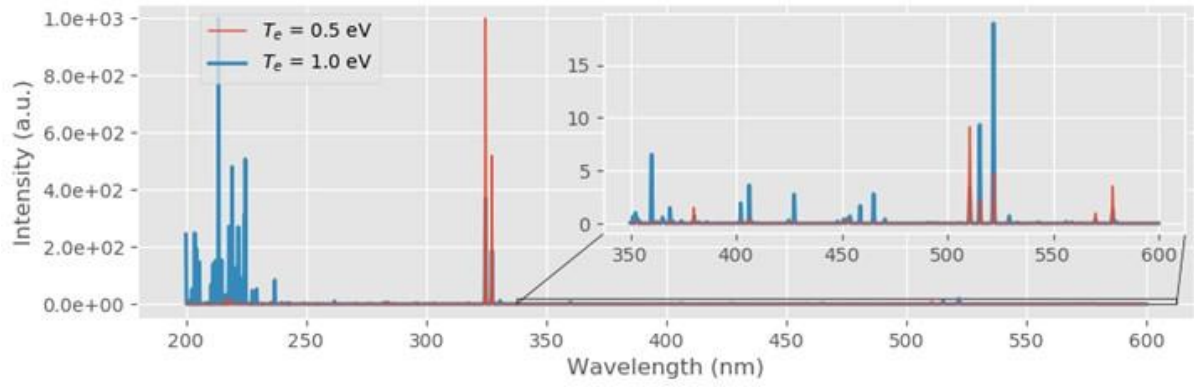
We got more intense emission in the shorter wavelength (UV) region at higher temperatures. This is because the higher energy states become more populated at higher temperatures, and this results in more intense UV emissions. Also, as a result of more ionizations, transitions of higher ionized species become intense at these temperatures.

The reported emission lines of LIP spectra of copper are 324.754 nm, 327.396 nm, 465.112 nm, 510.554 nm, 515.324 nm, 521.820 nm, and 578.213 nm; of zinc are 330.293 nm, 334.501 nm, and 472.215 nm, and 481.05 nm; and of tin are 270.65 nm, 283.997 nm, 286.331 nm, 300.913 nm, 303.411 nm and 326.233 nm [10,11]; and they are present in the simulated spectrum as well. Spectroscopic information on these lines are given in Table (1). There are also some lines which are closely packed and combined into a single line and are practically indistinguishable. Therefore, we can infer that simulations in AES give prior knowledge of intense characteristics emission of any samples.

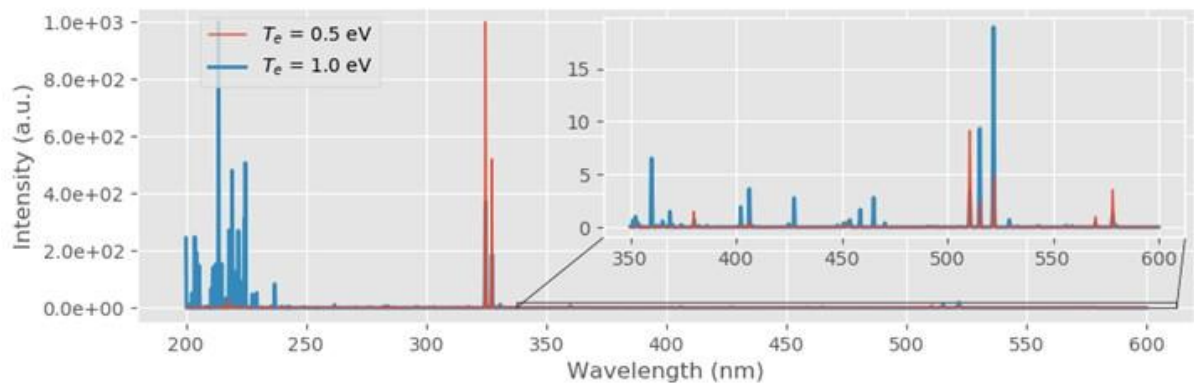
These simulated plasma spectra are useful for identifying possible lines we could observe in a real life spectrum. Then we could choose which lines are useful for the qualitative and quantitative analysis of multi-element samples in Laser-Induced Breakdown Spectroscopy (LIBS). In LIBS, observed intensities of known lines of the component elements are used to estimate plasma conditions and composition of the sample.



(a) Brass



(b) Bronze



(c) Admiralty Brass

Figure 1: Simulated spectrum of the three alloys at temperatures of 0.5 and 1 eV ($n_e = 10^{17} \text{ cm}^{-3}$ and $R = 1000$). An enlarged view of 350-600 nm region in inset. All spectra are normalised so that the strongest line have intensity 1000.

Table 1: Persistent lines that are observed in the spectra. Lines of a specific species will be present only if that is a component of the alloy considered.

Species	$\lambda(\text{nm})$	$A_{ul}(\text{s}^{-1})$	$E_l(\text{eV})$	$E_u(\text{eV})$	g_l	g_u
CuI	324.754	1.395e+08	0.00000	3.81669	2	4
CuI	327.396	1.376e+08	0.00000	3.78590	2	2
CuI	465.112	3.800e+07	5.07209	7.73703	10	8
CuI	510.554	2.000e+06	1.38895	3.81669	6	4
CuI	515.323	6.000e+07	3.78590	6.19118	2	4
CuI	521.820	7.500e+07	3.81669	6.19203	4	6
CuI	578.213	1.650e+06	1.64223	3.78590	4	2
ZnI	330.294	6.700e+07	4.02966	7.78233	3	3
ZnI	334.501	1.700e+08	4.07788	7.78335	5	7
ZnI	472.216	4.576e+07	4.02966	6.65451	3	3
ZnI	481.053	7.004e+07	4.07788	6.65451	5	3
SnI	270.650	6.600e+07	0.20976	4.78937	3	5
SnI	283.998	1.700e+08	0.42498	4.78937	5	5
SnI	286.331	5.400e+07	0.00000	4.32882	1	3
SnI	300.913	3.800e+07	0.20976	4.32882	3	3
SnI	303.411	2.000e+08	0.20976	4.29491	3	1
SnI	326.233	2.700e+08	1.06787	4.86725	5	3

The spectrum generated here is an ideal one, assuming ideal conditions. Simulating the exact spectrum requires more theoretical considerations and additional parameters like absorptions, broadening mechanisms etc. has to be considered. But this spectrum could be a useful tool to LIBS applications. It can be used to identify strong lines that could be present in the spectrum. It can also be used to choose which spectral window is to be sampled for specific conditions of the plasma, to get enough emission lines that are useful for analysis.

4. Conclusion

We have generated the plasma spectrum for three alloys of copper, tin and zinc under different plasma conditions. These spectra has many of the characteristic lines of the constituent species' and has shown a strong dependence on the plasma temperature. More intense lines were observed in the UV region as temperature increased. The spectrum simulated here is not a replica of an exact one. But it is a useful approximation under ideal conditions. Spectral lines and their intensities encodes the qualitative and quantitative information about the plasma. In LIBS, such a simulated spectrum can be used for identifying lines that are useful for the analysis. It can also be used to tell which spectral region has to be chosen for LIBS study, for a specific plasma condition.

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References

- [1] Capitelli M, Casavola A, Colonna G and De Giacomo A 2004 *Acta Part B: Atomic Spectroscopy* **59** 271–89
- [2] Cristoforetti G, Tognoni E and Gizzi L A 2013 *Spectrochimica Acta Part B: Atomic Spectroscopy* **90** 1–22

- [3] Cristoforetti G, De Giacomo A, Dell'Aglia M, Legnaioli S, Tognoni E, Palleschi V and Omenetto N 2010 *Spectrochimica Acta Part B: Atomic Spectroscopy* **65** 86–95
- [4] Kramida A and Ralchenko Y 1999 NIST Atomic Spectra Database, NIST Standard Reference Database 78
- [5] Kurucz R and Bell B 1995 Atomic Line Data *Atomic Line Data (R.L. Kurucz and B. Bell) Kurucz CD-ROM No. 23. Cambridge* **23**
- [6] Griem H R 1997 Principles of Plasma Spectroscopy, Cambridge Uni Press, Cambridge
- [7] Noll R 2012 Laser-Induced Breakdown Spectroscopy *Laser-Induced Breakdown Spectroscopy: Fundamentals and Applications* ed R Noll (Berlin, Heidelberg: Springer) pp 7–15
- [8] International Union of Pure and Applied Chemistry 1996 Compendium of Chemical Terminology, 2nd ed. (the "Gold Book") (Blackwell Scientific Publications, Oxford)
- [9] Weisstein E W Gaussian Function from MathWorld-A Wolfram Web Resource, [Online]. Available: <https://mathworld.wolfram.com/GaussianFunction.html> [2021, October 2].
- [10] Achouri M, Baba-Hamed T, Beldjilali S A and Belasri A 2015 *Plasma Phys. Rep.* **41** 758–68
- [11] Elhassan A, Giakoumaki A, Anglos D, Ingo G M, Robbiola L and Harith M A 2008 *Spectrochimica Acta Part B: Atomic Spectroscopy* **63** 504–11